Underground Activism Hub Project Progress Report

Introduction

Radar! Underground Activism Hub is a secure and private communication platform designed to facilitate activism and community organization. It prioritizes user privacy and security, ensuring that conversations remain confidential and inaccessible to unauthorized parties. The platform is inspired by the need for secure, decentralized communication tools that empower activists and communities to organize and collaborate safely.

Project Scope

The project aims to deliver a Minimum Viable Product (MVP) with core features that address the primary needs of users seeking secure communication channels. The key features include user authentication, chat rooms, real-time messaging, conversation handling, and media sharing.

Core Features (MVP)

1. **User Authentication & Security**
    - **No Sign-up, Login, and Logout**: Users can access the platform without creating accounts, enhancing anonymity.
    - **JWT Authentication**: JSON Web Tokens are used to authenticate users securely.
    - **Peer Verification**: Utilizes public-key cryptography to verify peers, ensuring that users can trust the identities of those they communicate with.
2. **Chat Rooms**
    - **Public Rooms**: Accessible to anyone with the room URL, featuring default random, unguessable room names. Custom names are allowed but may be more guessable.
    - **Private Rooms**: Require a password to join, enhancing privacy. Peers with mismatched passwords cannot connect, and no error is shown to maintain privacy.
    - **Peer Connection & Verification**: Room URLs can be shared securely using tools like Burner Note or Yopass. Public keys are displayed in the peer list for verification.
3. **Real-time Messaging**
    - Supports GitHub-flavored Markdown for rich text formatting and code syntax highlighting.

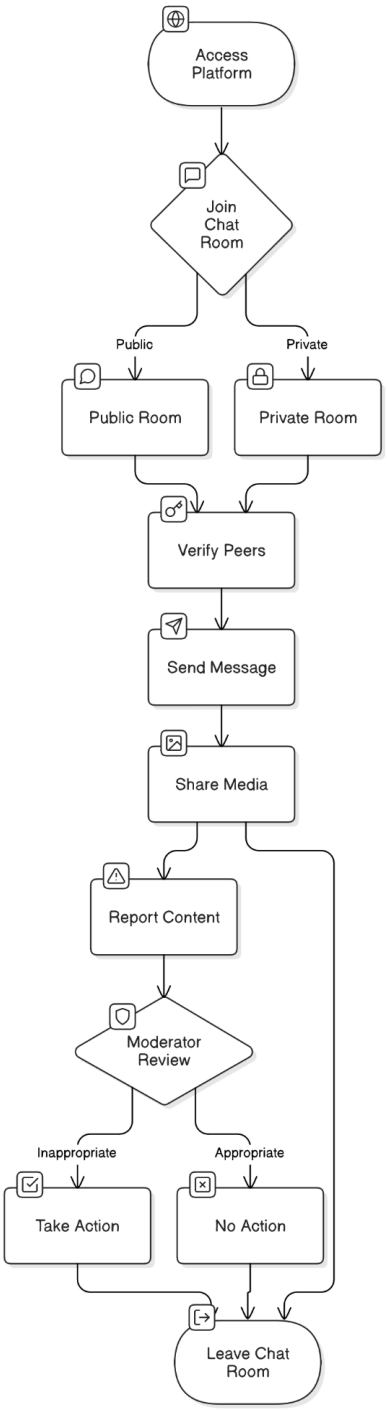
    - Messages are limited to 10,000 characters. Users can press Enter to send a message or Shift + Enter for a new line.
4. **Conversation Handling & Privacy**

    - Messages are stored in volatile memory only and are not retained permanently.
    - Conversations are completely erased when all peers leave the room.
    - Public rooms allow new peers to request the past 150 messages, while private rooms support real-time chats only, with no backfilling.
5. **Media Sharing**

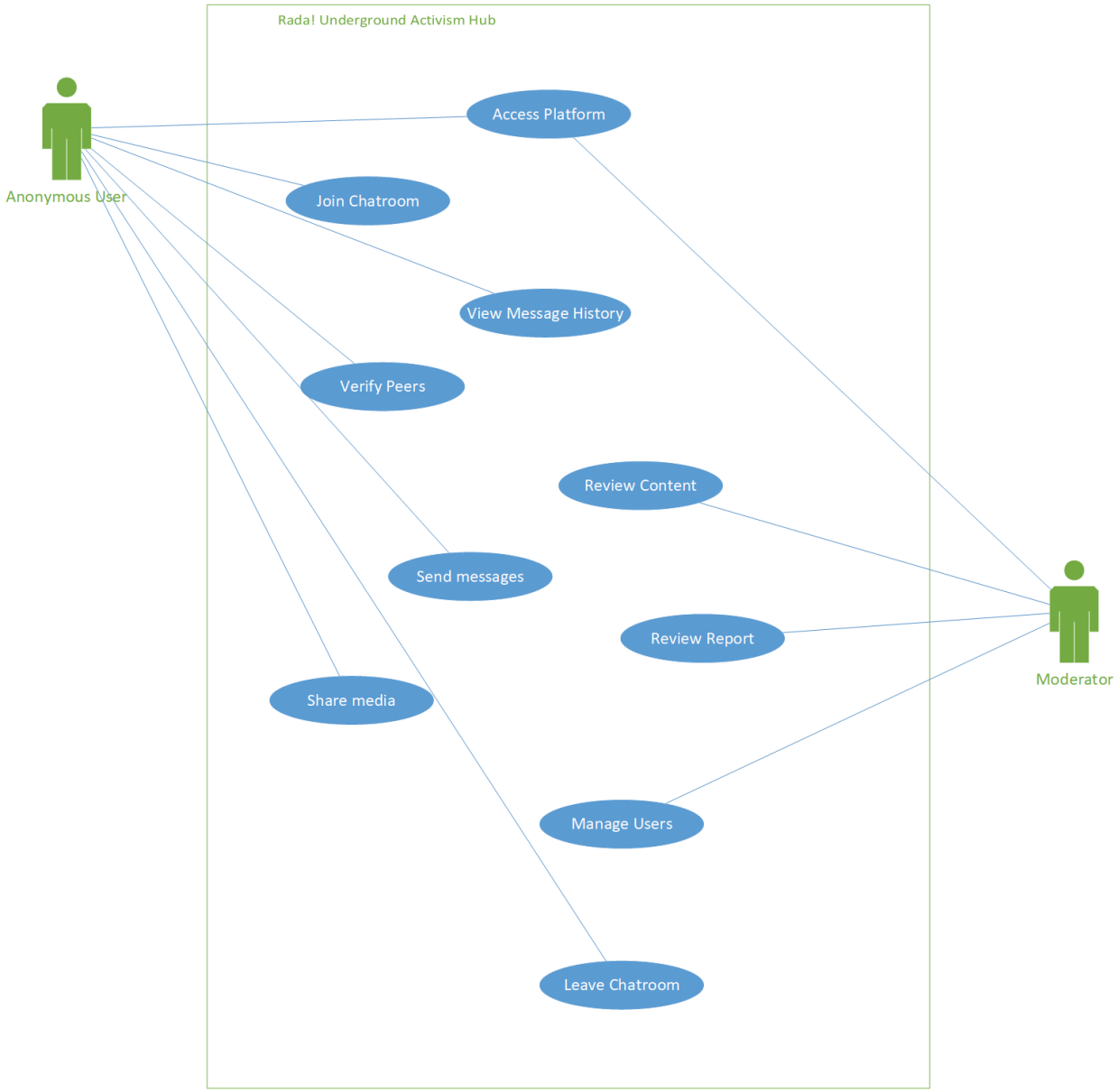
    - Users can upload and share images, audio, videos, and documents within the chat.
    - Preview functionality is available for images and videos.
    - File size restrictions are in place to optimize performance and ensure smooth user experience.

UML Diagrams

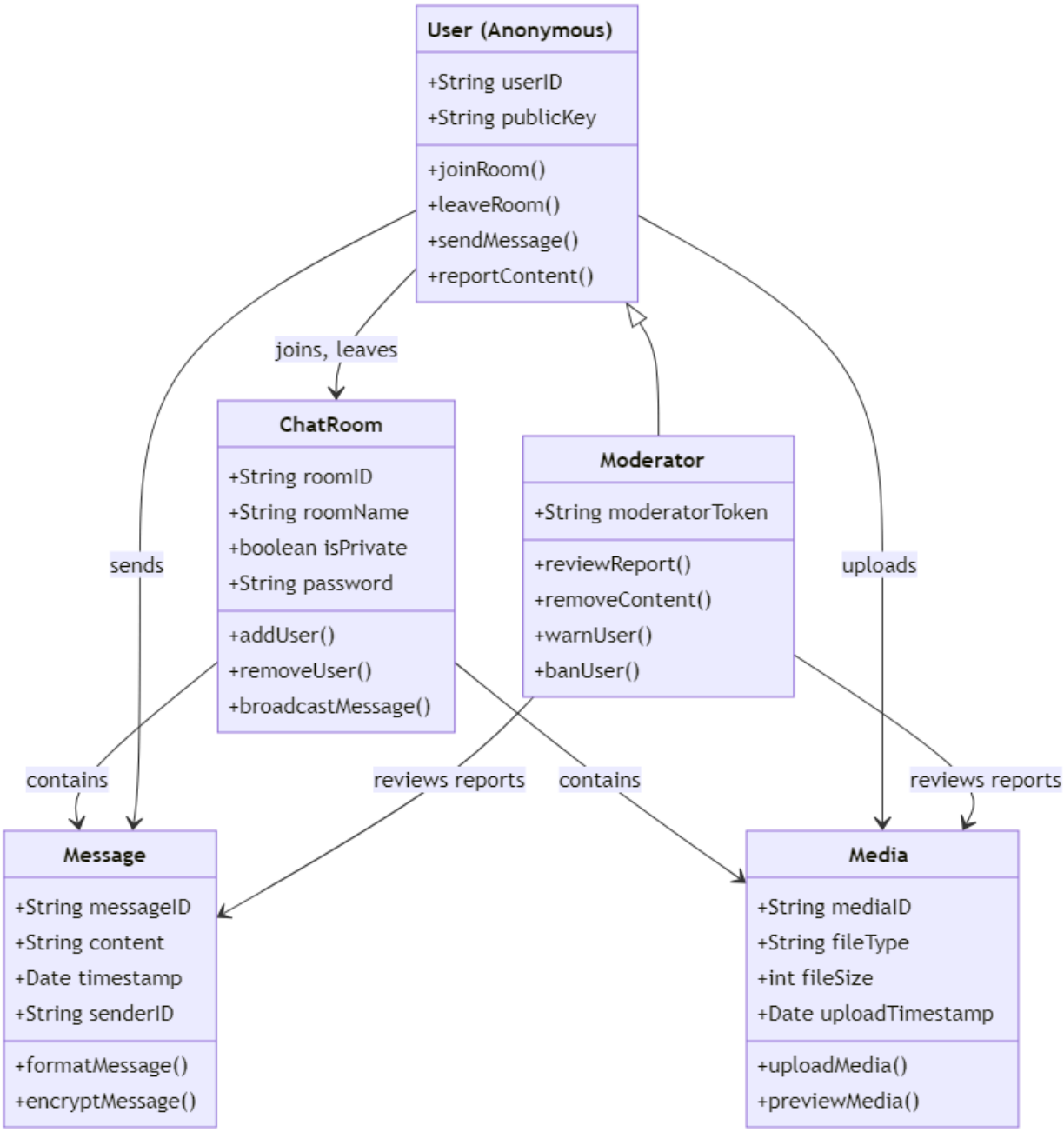
User Flow diagram



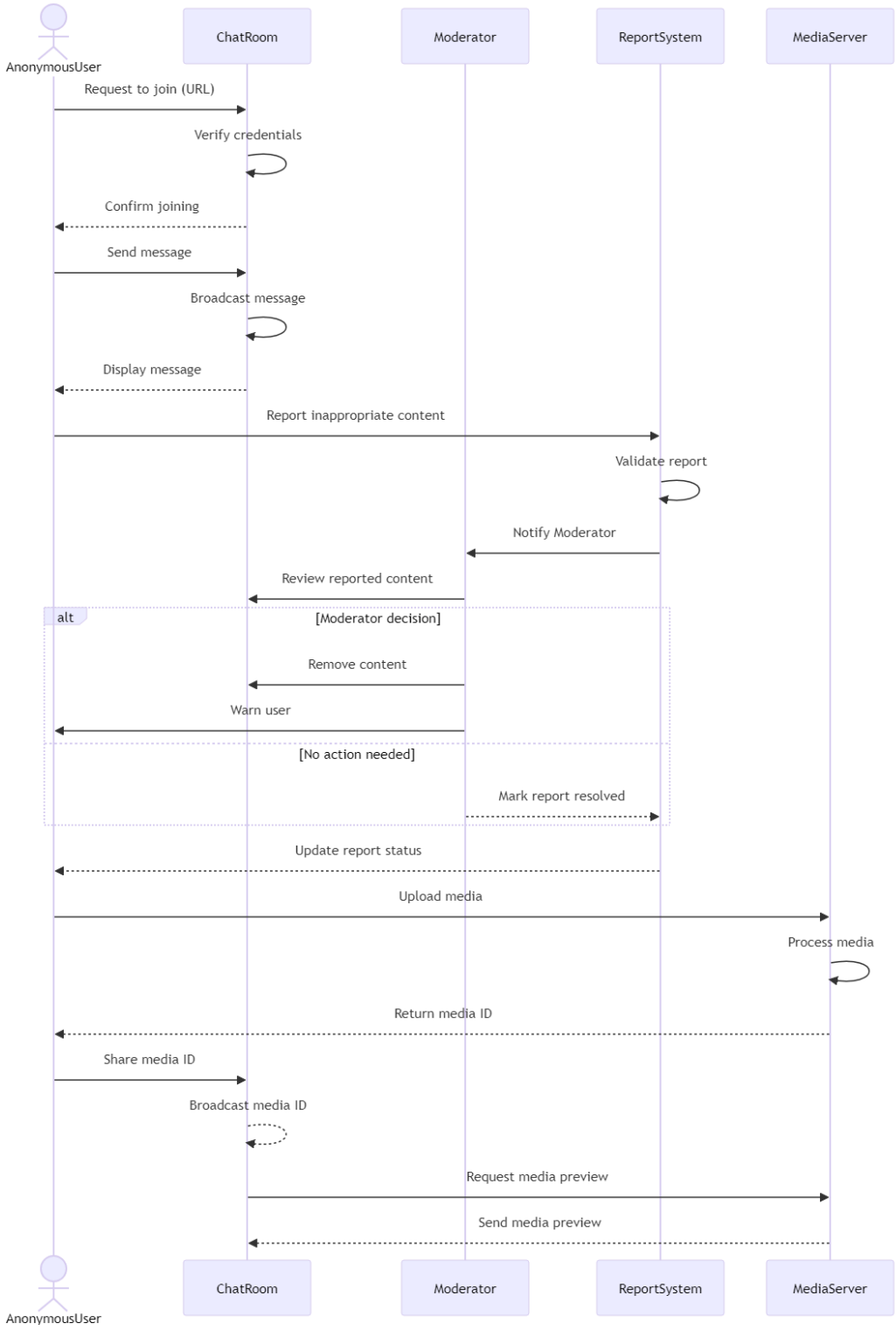
Use Case Diagram:



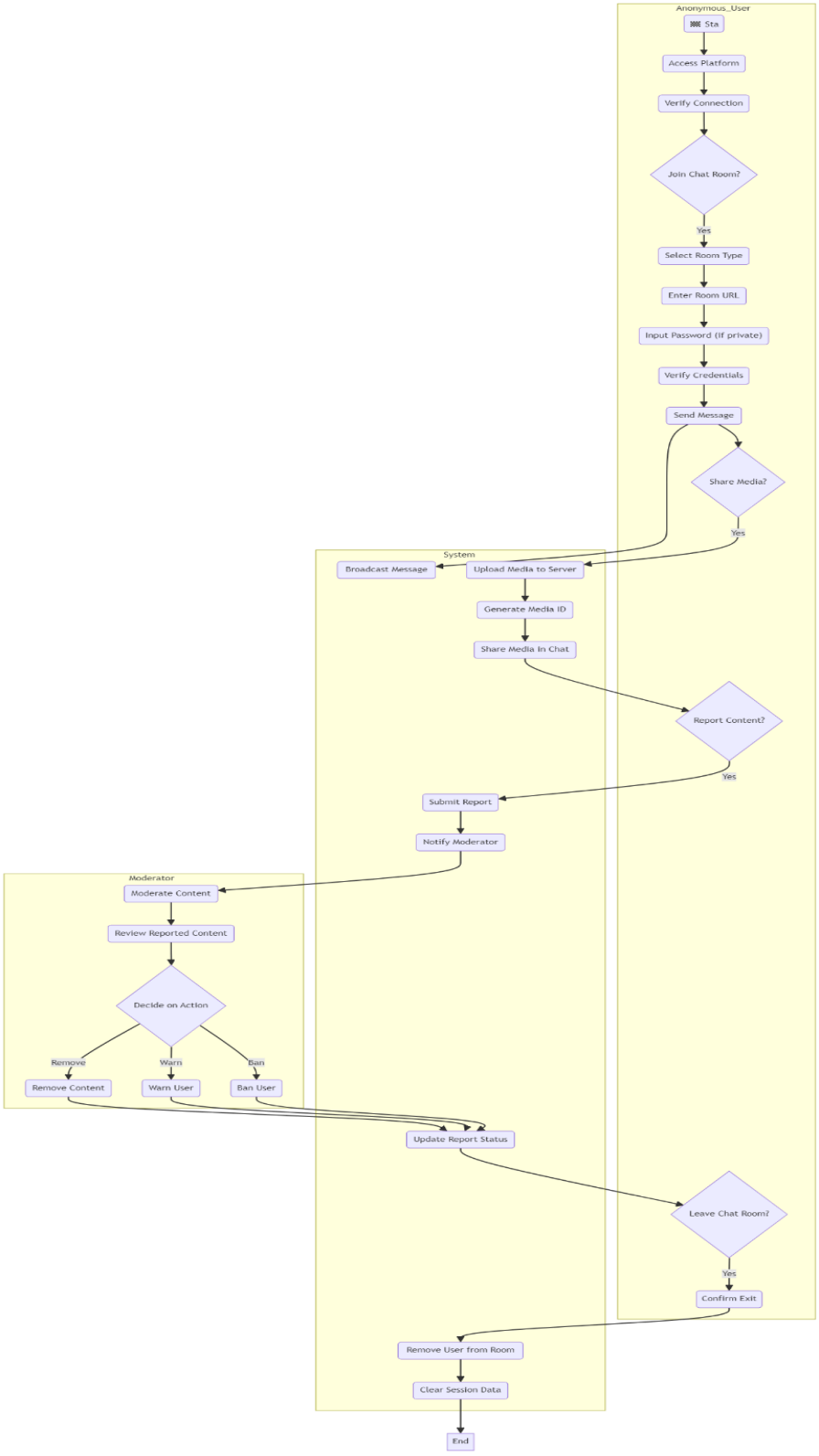
Class Diagram:



Sequence Diagram:



Activity Diagram:



Design Wireframes

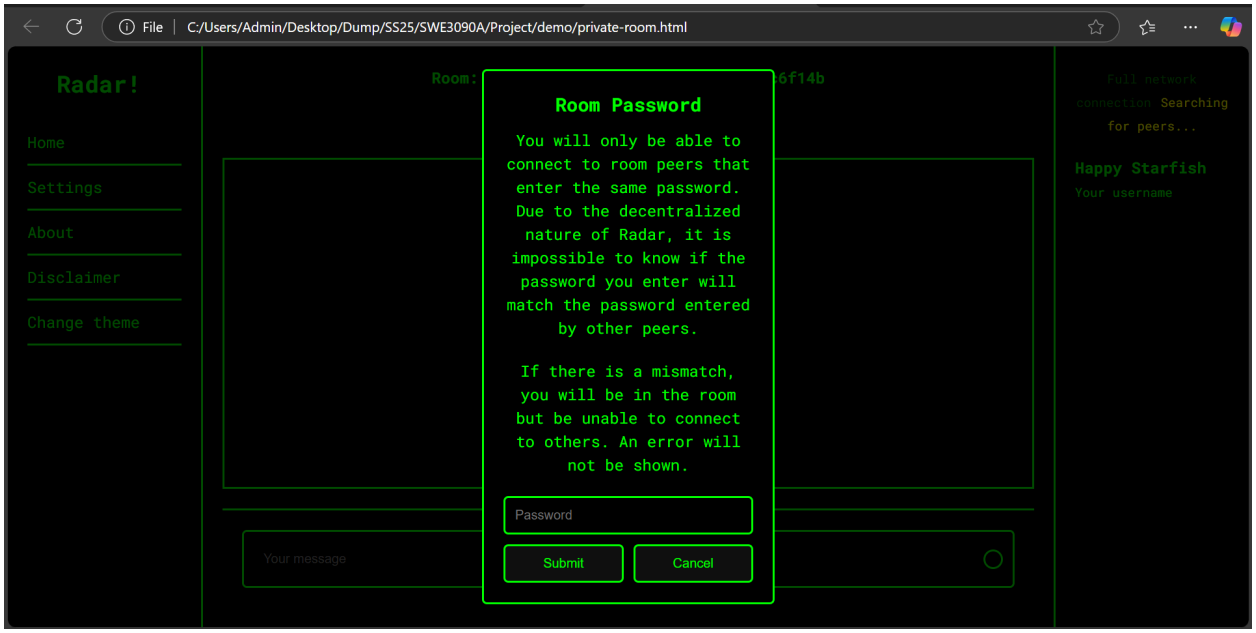
The design of the Underground Activism Hub is inspired by a Matrix-style theme, emphasizing a tech-focused and secure aesthetic. The wireframes depict the layout of the chat rooms, user authentication flows, and media-sharing interfaces. Each component is designed to be intuitive and efficient, supporting the platform's core functionalities.

Landing Page

Dashboard

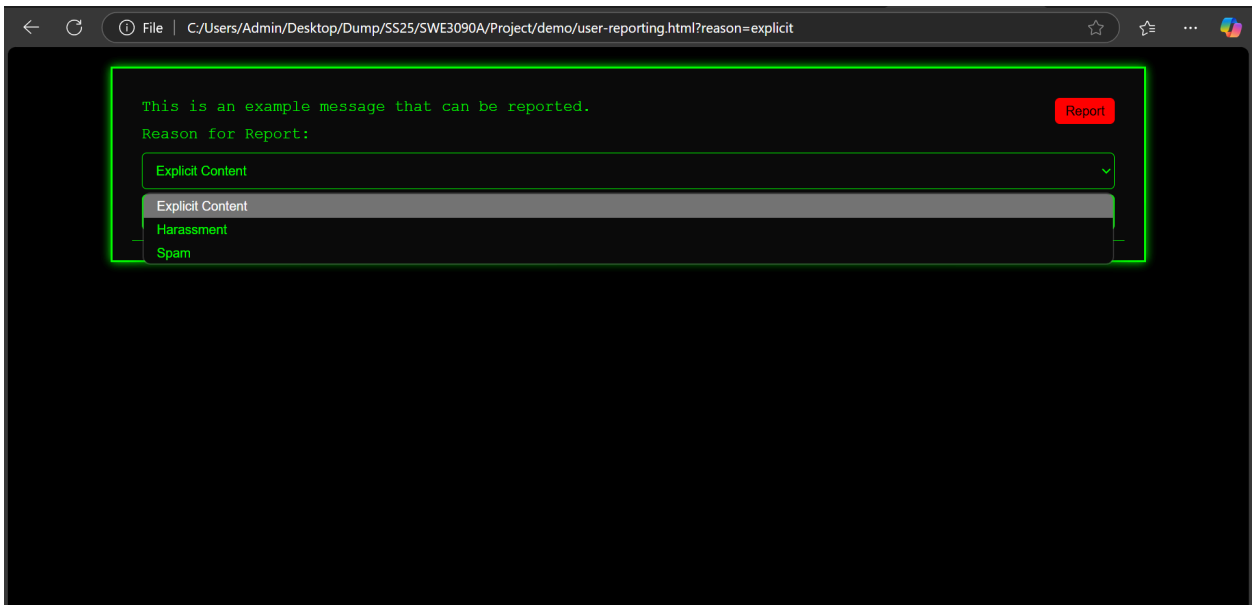
Public/Community/ Private chatroom layout

Private chat room password prompt

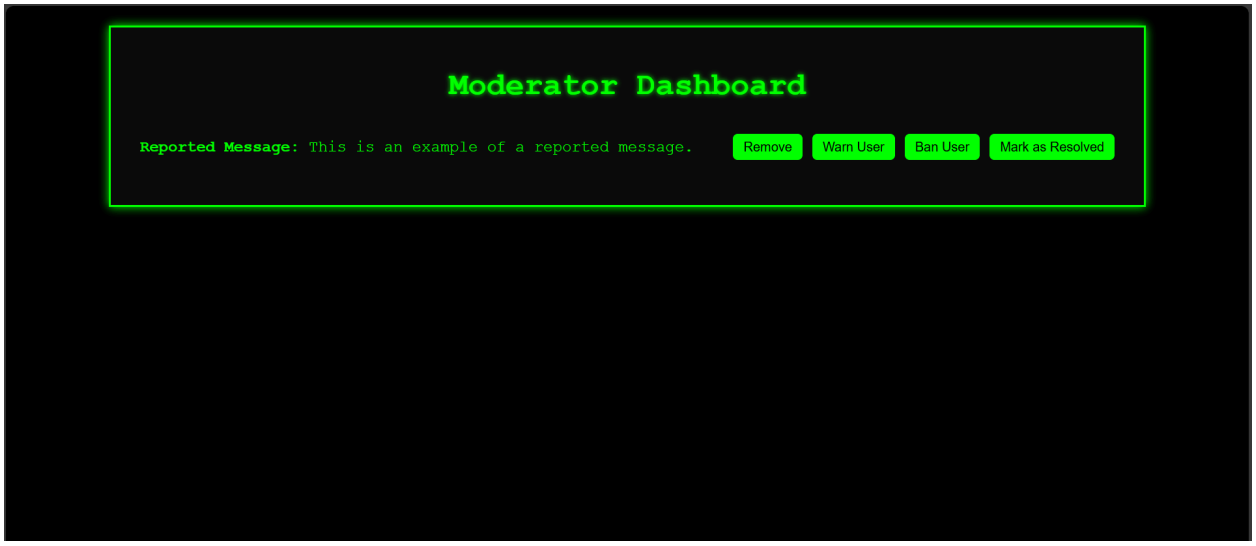


User Generated content moderation

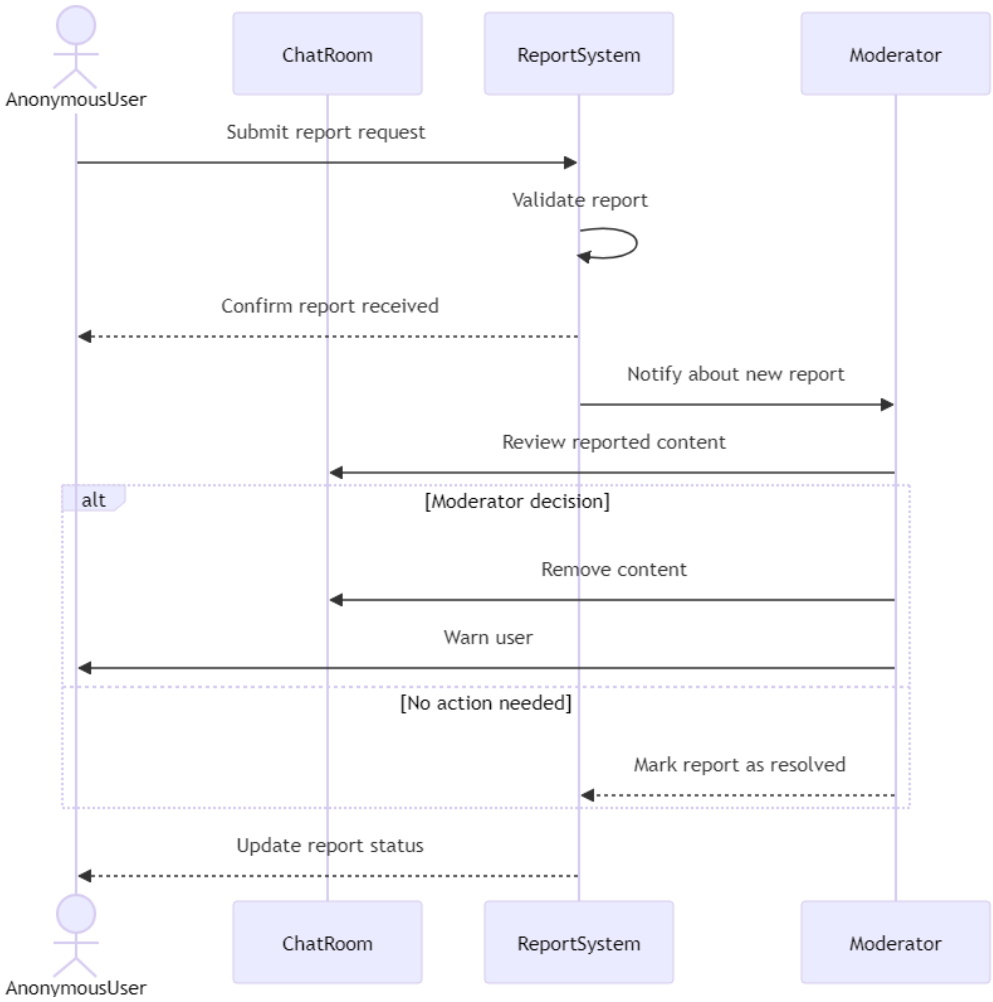
User reporting



Moderator dashboard



Sequence diagram for reporting mechanism



Challenges and Solutions

Redesign of UML Diagrams:

Challenge: The initial UML diagrams from my project proposal required redesigning to better align with the evolving project requirements and to address identified gaps in the system architecture.

Solution:

- **Reevaluation of Requirements**: Conducted a thorough review of the project requirements to ensure the UML diagrams accurately reflect the system's needs.
- **Iterative Design Approach**: Embraced an iterative design process, allowing for continuous refinement of the UML diagrams based on new insights and feedback.
- **Use of UML Tools**: Leveraged UML tools that support collaboration and version control to manage changes effectively.

Learning from Setbacks:

- Viewed the need for redesign as an opportunity to strengthen the design process. Analyzed the root causes of the setbacks and used this knowledge to improve future design activities.

Next Steps

- **Finalize UML Diagrams**: Continue refining the UML diagrams based on the latest requirements and stakeholder feedback.
- **Prototype Development**: Develop prototypes based on the finalized UML diagrams to validate the design and identify potential issues early in the development process.
- **Continuous Improvement**: Maintain an open approach to feedback and be prepared to make further adjustments as the project progresses.

Conclusion

The design phase of the Underground Activism Hub project has presented challenges that have been addressed through a systematic and collaborative approach. By embracing iterative design and engaging with stakeholders, the project is well-positioned to deliver a secure and user-friendly communication platform for activist communities.